HarAnsh Singh Gugnani

agugnani@andrew.cmu.edu | (913) 484-0655 | www.linkedin.com/in/hsgugnani

Education

Carnegie Mellon University, Pittsburgh, Pennsylvania **August 2024 – May 2028**
Bachelor of Science in Mechanical Engineering GPA: 3.62/4.0

- **Relevant Coursework:** Fundamentals of Programming and CS, Linear Algebra, Multivariable Calculus, Engineering Statistics, Manual Machining
- **In Progress:** Statics, Thermodynamics, Differential Equations
- **Upcoming:** Fluid Mechanics, 3D Mechanics, Numerical Methods, Dynamics

Skills

Engineering Tools: SolidWorks, Fusion 360, Inventor, MATLAB

Programming: Python, Arduino

Machining and Fabrication: CNC Machining, Manual Machining, MIG Welding, Laser Cutting, 3D Printing

Professional Experience

Carnegie Mellon University – CERLAB, Undergraduate Research Apprentice **May 2026 – Present**

- Designing a custom quadcopter frame in Inventor in collaboration with a PhD researcher for autonomous indoor inspection, integrating CERLAB's existing LiDAR, RGB-D camera, and NVIDIA Jetson Orin NX payload for GPS-denied environments.
- Reverted drone arm geometry to a sandwich plate design to eliminate adhesive fixation used in the previous build, with the goal of improving rotational stability and simplifying fabrication.
- Redesigned the propeller guard attachment to transfer impact loads into the carbon fiber arm rather than the 3D printed connector, and incorporated a TPU-based vibration isolation layer between the flight controller and frame.
- Currently transitioning the project from design into fabrication and flight testing to validate structural and vibration-isolation improvements.

Blue Valley CAPS, Mechanical Engineering Intern **August 2023 – May 2024**

- Rebuilt a salvaged electric go-kart with a team of 4 over 8 months as one of two primary welders; extended the frame 8 inches using welded steel members to accommodate the battery and motor assembly; completed vehicle ran under its own power and was driven by multiple testers.
- Machined an oversized axle from 0.750" to 0.625" to meet hub tolerances while preserving the 0.750" sprocket-side bore.
- Designed and 3D printed a custom sprocket retention gear and modified the motor-side fastener geometry to eliminate chain interference while maintaining clamping force.

Carnegie Mellon University, Pre-College Residential Advisor **June 2025 – August 2025**

- Supervised and supported 45 high school students in a residential academic program, coordinating with program staff to manage logistics and address student concerns.
- Fostered an inclusive and collaborative community while ensuring student safety and well-being.

Project Experience

Autonomous Arduino Car **August 2025 – December 2025**

- Programmed an Arduino-based vehicle integrating IR and ultrasonic sensors for line detection and automatic braking based on distance thresholds.
- Developed modular code for motion control and sensor data processing, achieving accurate path following across varied surface and lighting conditions.

Manual Machining Project — Tic Tac Toe Board **August 2025 – December 2025**

- Designed and fabricated a machined tic-tac-toe board with threaded playing pieces and a tapped base, turned on a manual lathe with external threads cut on each piece to mate with tapped holes in the board, maintaining ±0.005" dimensional tolerances across all components.
- Machined and tapped four steel support blocks and assembled the system to ensure proper fit and alignment.